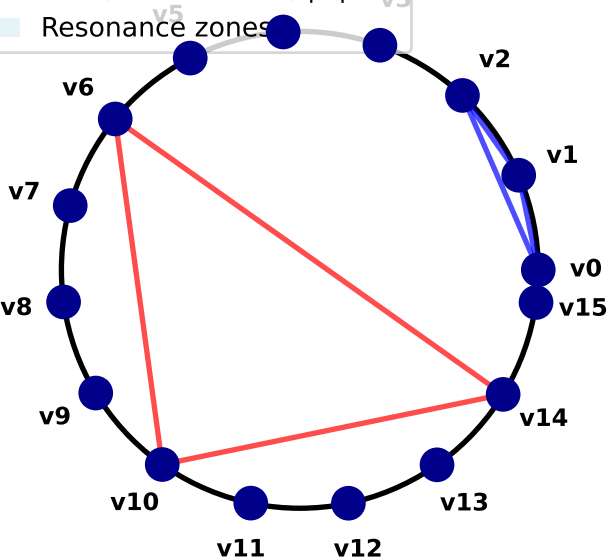


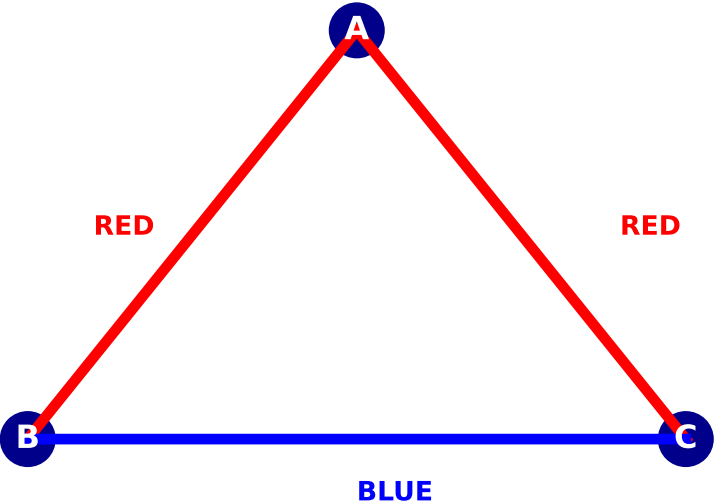
Vibrational Ramsey Coloring: $R\psi(5,5) \leq 16$

Vibrational Coloring on Circle
 $f_0 = 141.7001 \text{ Hz}$, $\epsilon = 0.037$

- Blue (resonant) $|\omega| < \epsilon$
- Red (non-resonant) $|\omega| \geq \epsilon$
- Resonance zones



Impossible Configuration
for Circular Distance



Constraints:
 $|\omega_a - \omega_b| \bmod f_0 \geq \epsilon$ (RED)
 $|\omega_a - \omega_c| \bmod f_0 \geq \epsilon$ (RED)
 $|\omega_b - \omega_c| \bmod f_0 < \epsilon$ (BLUE)
But: $|\omega_a - \omega_b| + |\omega_b - \omega_c| \geq |\omega_a - \omega_c|$
(Circular triangle inequality)
 $\Rightarrow$ CONTRADICTION for $\epsilon \ll f_0$